

Mathematics for GMP

Science

May, 2026

Short Title:	Mathematics for GMP	
Faculty	Science and Computing	
Department	Science	
Cluster	Mathematics and Physics	
Author	MBREEN	
Credits	10	Level: Introductory

Description of Module / Aims

This is an introductory course in mathematics and statistics. It assumed that the student has a knowledge of very basic mathematics only.

Programmes

MTHS-0019 Higher Certificate in Science in Good Manufacturing Practice and Technology 1 / 1 / M
(WD_SGMPT_C)

Indicative Content

- Mathematics fundamentals: fractions and percentages; units of measurement (S.I. units) & scientific notation; use of formulae; decimal places and significant digits; using a scientific calculator, including statistics mode; algebraic manipulation
- Statistics fundamentals: basic concepts; statistics and data defined; variables and variable measure (scale, ordinal, nominal). Variable type/role (experimental v observational, predictor v response). Descriptive v inferential statistics. Sampling (probability v non-probability sampling). Measures of central tendency (mean, median, mode) and spread (quartiles, standard deviation, range); plots -- histogram, stem and leaf, ogives, boxplot, bar chart, pie-chart, pareto diagrams
- The normal distribution; calculating probabilities and quartiles; applications, e.g. impacts on defective ppm of change in process parameters; assessing normality - probability plots
- Acceptance sampling; operating characteristics curves. the binomial distribution.
- Simple regression: scatter plots and graphical estimation of regression equation; prediction
- Software: using MS Excel; basic tabular calculations; Excel's statistical functions (mean, SD, normal, binomial, regression etc.); add-ins - Solver, data analysis; constructing / configuring charts in Excel

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Complete arithmetic calculations and simple percentage calculations.
2. Use fundamental concepts to solve problems in linear algebra.
3. Use formula and calculators to calculate measures of centre and spread.
4. Illustrate data using histograms, ogives, scatter plots and stem and leaf diagrams.
5. Complete simple regression and use regression equation for prediction.
6. Use software to carry out statistical calculations.

Learning and Teaching Methods

- Lectures.
- IT practicals.
- Reading recommended texts.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
<i>In-Class Assessment</i>	25%	1,2,3,4,5
<i>Assignment</i>	25%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- O' Reilly , J. *_Applied Statistics [__ebook__]*. Dublin: Statistical Solutions, 2014.

Supplementary Material

- Stroud, K. A. *_Foundation Mathematics_*. UK: Palgrave Macmillan, 2009.

Requested Resources

- Room Type: Computer Lab